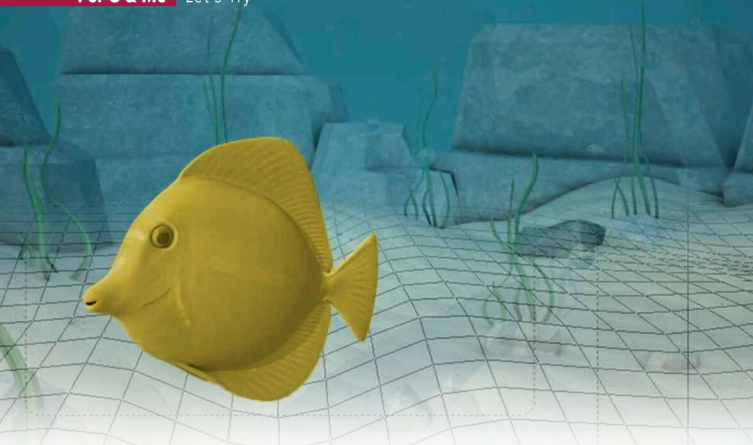




Learn to Animate in **BLENDER**

Part—3

This part of the series deals with creating controls for the fish and animating it.

Blender is great software for animation and is particularly suited for character animation. Following a process of modelling and creating vertex groups for complex models, and then auto-assigning bones using the same nomenclature as for the vertex groups, makes it very easy to set up a working character rig. The Armature tool in Blender is very advanced, yet quite simple to use even for novices. Setting up controls is also quite easy in Blender, compared to software like Maya or Max.

Create controls and add IK

Shift to the orthographic side view by pressing 3 and 5 on the numeric pad. Click just in front of the mouth of the fish. Press the space bar and *Add-Mesh-Cube*. Scale down the cube as shown in Figures 1 and 2.

You can create IK in Blender in various ways. The easiest is to first shift to Pose Mode. Select the control cube you just created.

Shift-select the head bone. Press *Ctrl+I*. A pop-up asks you whether you want to assign IK to the active object. Click *OK*. Go down to the Armature panel and set the chain length to 1.

Add similar cubes to create controls for the fins and the tail. Add IK for all the controls. Shift to the Animation UI and click the render boxes next to the cube icons in the outliner (*Restrict/Allow renderability*). This will disable rendering of the cubes (since in our renders we want only the goldfish to be displayed and not the controls). Figures 3 to 6 show the process of adding controls and IK, and Figure 7 shows how to toggle rendering of the cube.

Testing the controls

Next, it is time to test the controls. While in object mode, move around the cubes to test the deformations on the fish. If you find these are not as expected, shift to the weight painting mode and try to get rid of overlapping weights—